

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1536

Spearman City 100 kV
June 24, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1536. The Interconnection Customer proposes a 272.4 MW battery energy storage facility interconnecting at the Spearman City 100 kV substation in the Ohio area. The request was received on July 29, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 409 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 23 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$116,623,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Don-Hato Rey Gis 1403-1404 100 KV line	Dutton Pump-Celina Solar Project 1 Llc 100 KV	373.5	87	100.0
Don-Monroeville 100 KV	Dutton Pump-Celina Solar Project 1 Llc 100 KV line	300.2	113	100.0
Edes Tap 1-Hato ReyGis 1403-1404 100 KV	Dutton Pump-Celina Solar Project 1 Llc 100 KV Line	261.9	118	100.0
Spearman City-Edes Tap 1 100 KV	Dutton Pump-Celina Solar Project 1 Llc 100 KV	256.9	119	100.0
Dutton Pump-Celina Solar Project 1 Llc 100 KV Line	Don Substation-Monroeville 100 KV L	256.8	133	100.0
St Marys-Leyendecker 100 KV	Sherman St Substation-Eastmnt 100 KV L	197.0	147	61.2
Sylmar East-Celina Solar Project 1 Llc 100 KV Line	Celina Solar Project 1 Llc Substation-Welton Mohawk 100 KV	181.0	125	69.5
Sylmar East-Spearman City 100 KV L	Celina Solar Project 1 Llc-Welton Mohawk 100 KV line	180.3	122	69.5
Welton Mohawk-Spearman City 100 KV	Celina Solar Project 1 Llc-Sylmar E 100 KV	175.0	110	63.4
Welton Mohawk-Celina Solar Project 1 Llc 100 KV	Celina Solar Project 1 Llc-Sylmar East 100 KV	173.0	119	63.4
E Drainage Canal-Lenkurt 100 KV	St. Lawrence-James De Young 230 KV	154.1	134	9.7
Harper-West Olympia 100 KV Ckt 1	Harper-West Olympia 100 KV Cir. 2	135.5	147	11.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W Olympia-Harper 100 KV	West Olympia-Harper 100 KV Cir. 1	135.5	147	11.3
Glenciff-Lenkurt 100 KV line	Lenkurt-East Drainage Canal 100 KV	133.7	112	13.1
Beaverley-Geisel Monument 100 KV Line	St. Lawrence-James De Young 230 KV	116.6	187	12.3
Southwest Lima-Beaverley 100 KV L	St. Lawrence-James De Young 230 KV	115.4	199	12.3
R.P.Mone-Monroeville Substation 100 KV Line	Duttn Pump-Celina Solar Project 1 Llc 100 KV	114.8	104	37.5
Southwest Lima-Lenkurt 100 KV	James De Young-St. Lawrence 230 KV	114.5	210	12.3
Chimacum-Haviland 100 KV line	Dutton Pump-Celina Solar Project 1 Llc Substation 100 KV line	108.1	143	30.7
Haviland-Taconite Harbor Energy Ctr 100 KV Line	Celina Solar Project1 Llc-Dutton Pump 100 KV	107.9	142	30.7
Oak Plains-Eastmont 100 KV	Dutton Pump-Leyendecker 100 KV line	105.5	170	22.7
Dutton Pump-Leyendecker 100 KV	Eastmont-Sherman St 100 KV	103.2	256	69.8
Dixon Substation-Muskogee 100 KV	N. Lee-Muskogee 100 KV	102.0	146	9.5

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Don-Hato Rey Gis 1403-1404 100 KV	Rebuild line with higher-capacity conductor	\$1,533,000	100% (\$1,533,000)
Don-Monroeville 100 KV line	Rebuild line with higher-capacity conductor	\$15,534,000	100% (\$15,534,000)
Edes Tap 1-Hato ReyGis 1403-1404 100 KV	Rebuild line with higher-capacity conductor	\$8,888,000	100% (\$8,888,000)
Spearman City-Edes Tap 1 100 KV L	Rebuild line with higher-capacity conductor	\$17,014,000	100% (\$17,014,000)
Dutton Pump-Celina Solar Project 1 Llc 100 KV Line	Rebuild line with higher-capacity conductor	\$3,387,000	100% (\$3,387,000)
St Marys-Leyendecker 100 KV line	Rebuild line with higher-capacity conductor	\$694,000	100% (\$694,000)
Sylmar East-Celina Solar Project 1 Llc 100 KV Line	Rebuild line with higher-capacity conductor	\$2,624,000	100% (\$2,624,000)
Spearman City-Sylmar East 100 KV	Rebuild line with higher-capacity conductor	\$3,158,000	100% (\$3,158,000)
Spearman City-Welton Mohawk 100 KV	Rebuild line with higher-capacity conductor	\$5,015,000	100% (\$5,015,000)
Celina Solar Project 1 Llc-Welton Mohawk 100 KV	Rebuild line with higher-capacity conductor	\$6,576,000	100% (\$6,576,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E Drainage Canal-Lenkurt 100 KV	Rebuild line with higher-capacity conductor	\$11,530,000	100% (\$11,530,000)
Harper-West Olympia 100 KV Ckt 1	Rebuild line with higher-capacity conductor	\$2,086,000	100% (\$2,086,000)
W Olympia-Harper 100 KV	Rebuild line with higher-capacity conductor	\$3,711,000	100% (\$3,711,000)
Glenclyff-Lenkurt 100 KV line	Rebuild line with higher-capacity conductor	\$4,313,000	100% (\$4,313,000)
Beaverley-Geisel Monument 100 KV	Rebuild line with higher-capacity conductor	\$2,668,000	100% (\$2,668,000)
Southwest Lima-Beaverley 100 KV	Rebuild line with higher-capacity conductor	\$2,701,000	100% (\$2,701,000)
R.P.Mone-Monroeville 100 KV	Rebuild line with higher-capacity conductor	\$6,395,000	100% (\$6,395,000)
Southwest Lima-Lenkurt 100 KV line	Rebuild line with higher-capacity conductor	\$5,652,000	100% (\$5,652,000)
Chimacum-Haviland 100 KV	Reconductor line	\$1,358,000	100% (\$1,358,000)
Taconite Harbor Energy Ctr-Haviland 100 KV	Reconductor line	\$2,088,000	100% (\$2,088,000)
Oak Plains-Eastmont 100 KV L	Reconductor line	\$4,905,000	100% (\$4,905,000)
Dutton Pump-Leyendecker 100 KV	Reconductor line	\$2,669,000	100% (\$2,669,000)
Muskogee-Dixon 100 KV	Reconductor line	\$2,124,000	100% (\$2,124,000)
	Total allocated to project		\$116,623,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.